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A generalized finite difference method for solving Stokes interface problems

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ABSTRACT

In this paper, a new scheme is proposed to solve the Stokes interface problem. The scheme turns the Stokes interface problem into two coupled Stokes non-interface subproblems and adds a mixed boundary condition to overcome the numerical pressure oscillation. Since the interface becomes the boundary of the subproblems, the scheme has the advantage to deal with the interface problem with complex geometry. Furthermore, a generalized finite difference method (GFDM) is adopted to solve the coupled Stokes non-interface subproblems. The GFDM is developed from the Taylor series expansions and moving-least squares approximation. Due to the flexibility of the GFDM, it is convenient to handle the complex boundary conditions that appeared in the proposed scheme. The numerical examples verify the accuracy and stability of the GFDM to solve the Stokes interface problem with the mixed boundary conditions. Moreover, for some given numerical examples, the proposed scheme is more accurate than the classical formula of the pressure Poisson equation, especially in terms of pressure.

1. Introduction

The Stokes interface problem comes from the fluid-structure interaction (FSI) problem [1] and the two-phase incompressible flow [2]. In the two-phase incompressible flow, the density and viscosity are discontinuous at the interface between two different fluids. Therefore, when dealing with the two-phase incompressible flow, the jumping condition of flux on the interface should be considered. The Stokes interface problem is considered to be a reasonable first step to study the low Reynolds number two-phase incompressible flow.

The discontinuity of the viscosity coefficient on the interface is one of the challenges for solving the interface problem. Up to now, several numerical strategies had been presented to deal with these challenges. A major idea was to normalize the discontinuities [3]. However, it will reduce the accuracy and needs fine meshes nearby the interface areas. To avoid this difficulty, Buscaglia et al. [4] established a finite element model for the pressure, which simulates the jump of the normal component of the surface force under the assumption that the tangential component of the surface force is continuous.

Hansbo et al. [5] proposed a cut FEM to solve the Stokes interface problem, which weakly enforces the jumping condition on the interface with a weighted coefficient in the Nitsche's numerical flux. Adjerid et al. [6] provided a discontinuous immersive (IFEM) based on Cartesian grids for the Stokes interface problem. Sun [7] studied a distributed Lagrange multiplier/fictitious domain (DLM/FD) FEM for the Stokes/elliptic interface problem and developed a mixed finite element discretization to solve the DLM/FD method.

Among the FEM to solve the Stokes interface problem, the XFEM is another popular choice. Chessa et al. [8] applied an XFEM with arbitrary interior discontinuous gradients to the problem of two-phase immiscible flows. Reusken [9] improved the XFEM space by deleting finite element basis functions with fewer supports. Fries et al. [10] proposed the generalized FEM and XFEM for the problem with unsmooth solutions. Cattaneo et al. [11] combined the standard mixed finite element spaces with simple enrichment strategies to obtain a stable XFEM for the Stokes interface problem. In the absence of theoretical analysis, Capatina et al. [12] illustrated the approximate value of the Stokes interface problem numerically. Wang et al. [13] proposed a nonconforming Nitsche's XFEM for the Stokes interface problem. In addition, there exist other grid methods to solve the fluid interface problem, such as the immersion interface method (IIM) [14–16] and the boundary element method [17], etc.

In recent years, meshless methods have also been used to solve the Stokes interface problem. Hessari [18] proposed and analyzed the 1st-order system least square method of Stokes equations with discontinuous viscosity coefficients and singular forces along with the

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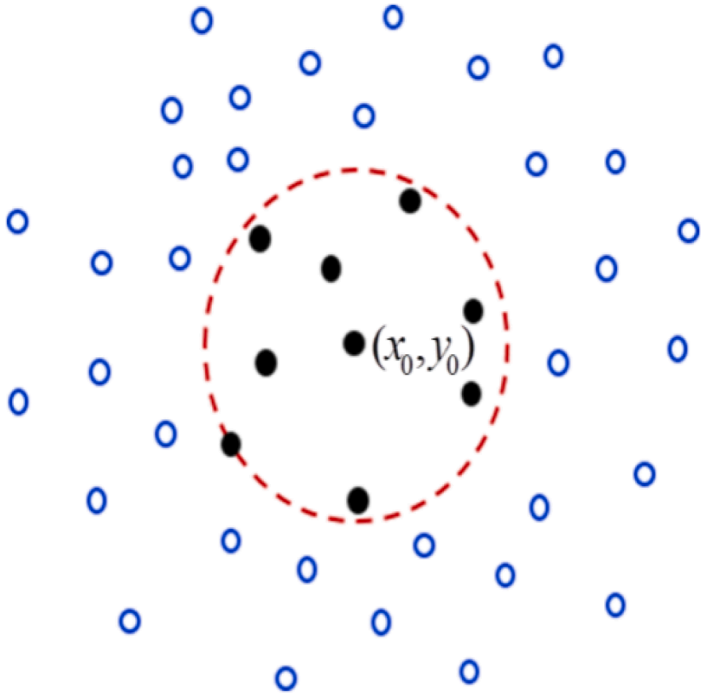


Fig. 1. The star of a central node (x_0,y_0) for a 2D case.

interface. Hessari [19] used the pseudo-spectral least squares method to approximate the two-dimensional Stokes interface problems. Ahmad et al. [20] proposed the local radial basis function collocation method to solve the Stokes interface problem numerically. The advantage of meshless methods to solve the interface problem is that they can flexibly handle complex computational domains and the interface condition.

In this paper, we use a meshless method named generalized finite difference method (GFDM) to solve the Stokes interface problem. The idea of GFDM was initially proposed by Lizska and Orkisz [21, 22] in the 1980s, and improved by Benito, Urena and Gavete [23–27]. The GFDM is based on the Taylor series expansions of multivariate functions and the weighted moving least-squares method to construct a numerical discretization scheme. Its main idea is that the derivatives of unknown variables can be expressed as linear combinations of nearby nodal values. Benito et al [23,25] proposed the GFDM formula for second-order two-dimensional partial differential equations. The influence of key parameters, including form of nodal distribution, weight functions, and shape of the supporting domain, has been well studied and can be seen as the important information for using GFDM. Recently, GFDM has been applied to many fields, such as some ill-posed inverse problems [28, 29], the elliptic interface problem [30], the elasticity problem [31, 32] and the Stokes problem [33].

The present work aims to use the GFDM to solve the Stokes interface problem. We divide the Stokes interface problem into two coupled Stokes non-interface subproblems. The coefficient matrix formed by these subproblems is sparse. We regard the interface as the boundary of each subproblem. Therefore, the subproblems only involve forcing boundary conditions at the boundary nodes. In addition, to overcome the pressure oscillation we adopt the mixed boundary condition proposed in [33] and provide a new alternative formulation of the Stokes interface problem. Furthermore, the GFDM uses the linear summation of nearby function values to approximate the derivative of unknown variables, so the new formulation is convenient to be solved by the GFDM.

This paper is organized as follows: the Stokes interface problem is described in Section 2. Section 3 shows the numerical procedure of the GFDM. The discrete scheme of the GFDM for the Stokes interface problem is shown in Section 4. In Section 5, some numerical examples are provided to verify the accuracy and stability of the proposed method. Finally, Section 6 draws some conclusions from this work.

2. Stokes interface problems

The Stokes interface problem models two incompatible and incompressible fluids on a bounded domain Ω. The two-dimensional (2D) problem finds the fluid velocity vector $\boldsymbol{u} = (u_1,u_2)$ and the fluid pressure p such that:

$$-\mu\Delta\boldsymbol{u} + \nabla p = \boldsymbol{f}, in\Omega, \tag{1}$$

$$\nabla\cdot\boldsymbol{u} = 0, in\Omega, \tag{2}$$

$$\boldsymbol{u} = \boldsymbol{g}, on\partial\Omega, \tag{3}$$

$$[\boldsymbol{u}]_\Gamma = \boldsymbol{\varphi}, on\Gamma, \tag{4}$$

$$[\sigma(\boldsymbol{u},p)\cdot\boldsymbol{n}]_\Gamma = \boldsymbol{\psi}, on\Gamma, \tag{5}$$

where $\partial\Omega$ is the boundary of the computational domain Ω, $\boldsymbol{f} = (f_1,f_2)$ is the vector of internal loads, $\boldsymbol{g} = (g_1,g_2)$ is the Dirichlet-type boundary condition of velocity vector $\boldsymbol{u}$, and the symbol $\boldsymbol{n} = (n_x,n_y)$ denotes the unit outward normal to the interface Γ. Throughout the paper, we assume the location and the shape of the interface Γare known to simplify the interface problem. In addition, $\sigma(\boldsymbol{u}, p) = -p\boldsymbol{I} + \mu\nabla\boldsymbol{u}$ is the stress tensor, there $\boldsymbol{I}$ is the unit matrix. The interface conditions are defined by Eqs. (4)–(5) with right-hand functions $\boldsymbol{\phi} = (\phi_1,\phi_2)$ and $\boldsymbol{\psi} = (\psi_1,\psi_2)$.

And μ is the piecewise constant viscosity coefficient:

$$\mu = \begin{cases} \mu^+, (x,y) \in \Omega^+, \\ \mu^-, (x,y) \in \Omega^-. \end{cases} \tag{6}$$

Here domain Ω^+ and Ω^- are sub-domains of the domain $\Omega = \Omega^+ \cup \Omega^-$, which are separated by an interface $\Gamma = \partial\Omega^+ \cap \partial\Omega^-$.

If the Stokes interface problem (1)-(5) is solved directly by numerical method, it usually leads to the numerical pressure oscillation. To overcome it, we adopt a new simple formulation named the mixed boundary condition (MBC) formulation of the Stokes problem proposed in [33]. It multiplies the unit outward normal vector at the boundary $\partial\Omega$ by the governing Eq. (1). The MBC formulation can also be applied for the Stokes interface problem. In addition, we add a new interface condition to get an alternative formulation of Eqs. (1)–(5) as follows:

$$-\mu\Delta\boldsymbol{u} + \nabla p = \boldsymbol{f}, in\Omega, \tag{7}$$

$$\nabla\cdot\boldsymbol{u} = 0, in\Omega, \tag{8}$$

$$\boldsymbol{u} = \boldsymbol{g}, on\partial\Omega, \tag{9}$$

$$-\mu\Delta\boldsymbol{u}\cdot\boldsymbol{n}_b + \nabla p\cdot\boldsymbol{n}_b = \boldsymbol{f}\cdot\boldsymbol{n}_b, on\partial\Omega, \tag{10}$$

$$[\boldsymbol{u}]_\Gamma = \boldsymbol{\varphi}, on\Gamma, \tag{11}$$

$$[\sigma(\boldsymbol{u}, p)\cdot\boldsymbol{n}]_\Gamma = \boldsymbol{\psi}, on\Gamma, \tag{12}$$

$$-\mu\Delta\boldsymbol{u}\cdot\boldsymbol{n} + \nabla p\cdot\boldsymbol{n} = \boldsymbol{f}\cdot\boldsymbol{n}, on\Gamma, \tag{13}$$

where $\boldsymbol{n}_b = (n_{bx}, n_{by})$ is the unit outward normal vector on the boundary $\partial\Omega$. Eq. (13) is obtained by multiplying the Eq. (1) with the unit outward normal $\boldsymbol{n}$ on the interface.

This form makes no change in the governing equations, which is different from the classical pressure Poisson equation (PPE) form:

$$\begin{cases} -\mu\Delta\boldsymbol{u} + \nabla p = \boldsymbol{f}, in\Omega \\ \Delta p = \nabla\cdot\boldsymbol{f}, in\Omega \\ \boldsymbol{u} = \boldsymbol{g}, on\partial\Omega \\ \nabla\cdot\boldsymbol{u} = 0, on\partial\Omega \\ [\boldsymbol{u}]_\Gamma = \boldsymbol{\varphi}, on\Gamma \\ [\sigma(\boldsymbol{u}, p)\cdot\boldsymbol{n}]_\Gamma = \boldsymbol{\psi}, on\Gamma \\ \nabla\cdot\boldsymbol{u} = 0, on\Gamma \end{cases}, \tag{14}$$

Obviously, Eq. (14) requires that the vector $\boldsymbol{f}$ is 1st-order differentiable and pressure p is 2nd-order differentiable. But these requirements do not appear in the MBC form.

3. The generalized finite difference method (GFDM)

The GFDM is a typical collocation meshless method. It is based on the Taylor series expansions of multivariate functions and the weighted moving least-squares method to construct a numerical discretization scheme. The main idea of the GFDM is to approximate the derivatives of an unknown variable at every given node as linear combinations of function values of its neighboring nodes.

For a given node (x_0,y_0), the m nearest nodes $(x_k,y_k)(k = 1, 2, \cdots, m)$ can be found. The node (x_0,y_0) and its nearest m nodes are collected to form a star, which is shown in Fig. 1. From previous studies [34–37], accurate and stable numerical solutions can be obtained when m is larger than 10 in the 2D case. The effect of the parameter m will be tested in Section 5. When the star of the central node (x_0,y_0) is formed, the Taylor series is used to expand the function inside the star as follows:

$$u_k = u_0 + h_k\frac{\partial u_0}{\partial x} + l_k\frac{\partial u_0}{\partial y} + \frac{1}{2}\left[h_k^2\frac{\partial^2 u_0}{\partial x^2} + l_k^2\frac{\partial^2 u_0}{\partial y^2} + 2h_k l_k\frac{\partial^2 u_0}{\partial x\partial y}\right] + \cdots, \quad k = 1, 2, \cdots m. \tag{15}$$

Where $u_k = u(x_k,y_k)$ is the functional value at nodes $(x_k,y_k)(k = 1, 2, \cdots, m)$, $\frac{\partial u_0}{\partial x}$, $\frac{\partial u_0}{\partial y}$ and $\frac{\partial^2 u_0}{\partial x^2}$, $\frac{\partial^2 u_0}{\partial y^2}$, $\frac{\partial^2 u_0}{\partial x\partial y}$ denote the first- and second-order derivative of $u(x)$ at the node (x_0,y_0). h_k and l_k denote the distance between nodes (x_k,y_k) and central node (x_0,y_0) along the x and y directions:

$$h_k = x_k - x_0, \quad k = 1, 2, \cdots m, \tag{16}$$

$$l_k = y_k - y_0, \quad k = 1, 2, \cdots m. \tag{17}$$

By truncating the Taylor series after the second-order (2nd-order) derivatives, we define a residual functional $B(u)$ as follows:

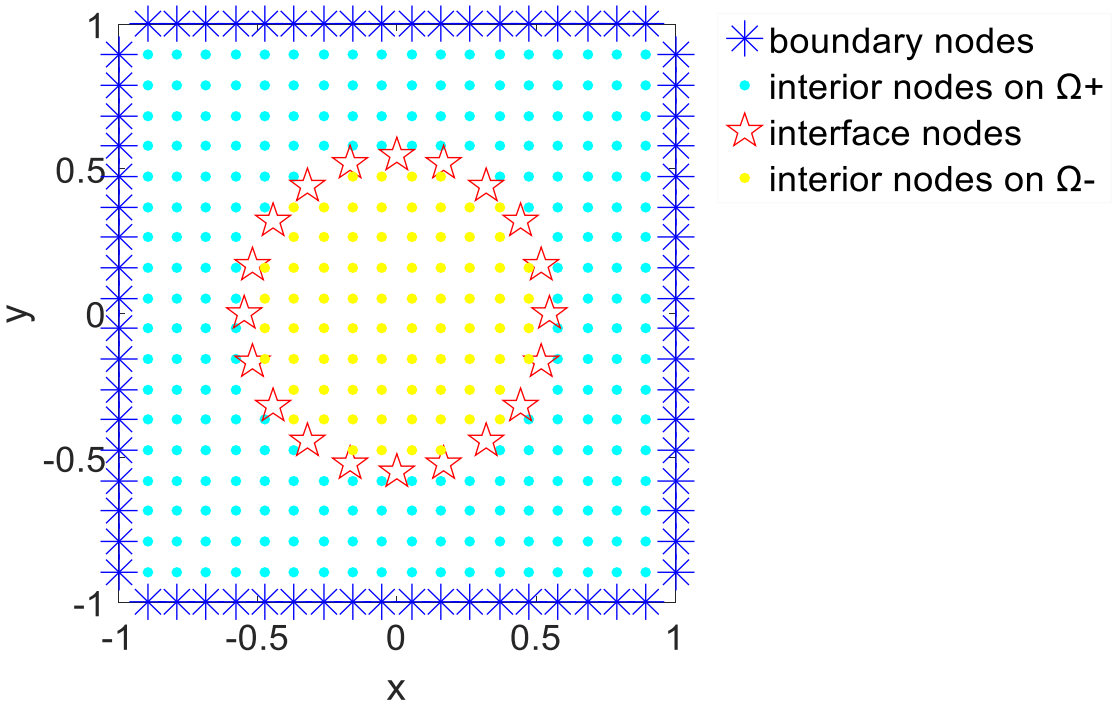


Fig. 2. The collocation points with $N_x = 20$ for Example 1.

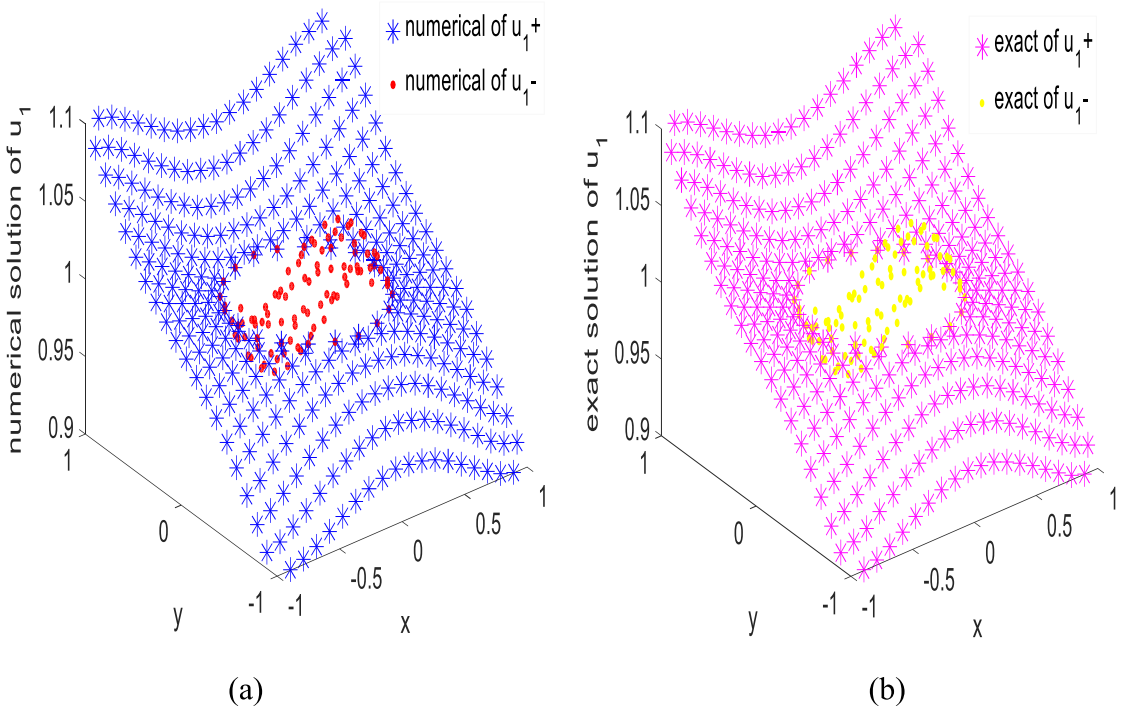


Fig. 3. The (a) numerical and (b) exact solutions of u_1 with $N_x = 20$ for Example 1.

$$B(u) = \sum_{k=1}^{m}\left[\left(u_0 - u_k + h_k\frac{\partial u_0}{\partial x} + l_k\frac{\partial u_0}{\partial y} + \frac{h_k^2}{2}\frac{\partial^2 u_0}{\partial x^2} + \frac{l_k^2}{2}\frac{\partial^2 u_0}{\partial y^2} + h_k l_k\frac{\partial^2 u_0}{\partial x\partial y}\right)\omega_k\right]^2, \tag{18}$$

where $\omega_k (k = 1, 2, \cdots, m)$ is the weighting coefficient at nodes (x_k, y_k). There are various choices for the weighting functions, such as potential function and cubic spine [23–26,38–39], etc. In our computations, we adopt the weight function as follows:

$$\omega_k = \begin{cases} 1 - 6\left(\frac{d_k}{d_m}\right)^2 + 8\left(\frac{d_k}{d_m}\right)^3 - 3\left(\frac{d_k}{d_m}\right)^4, & d_k \le d_m, \\ 0, & d_k \ge d_m. \end{cases} \tag{19}$$

Where $d_k = \sqrt{(x_k - x_0)^2 + (y_k - y_0)^2}$ denotes the distance between nodes (x_k, y_k) and central node (x_0, y_0), and d_m is the maximum value of

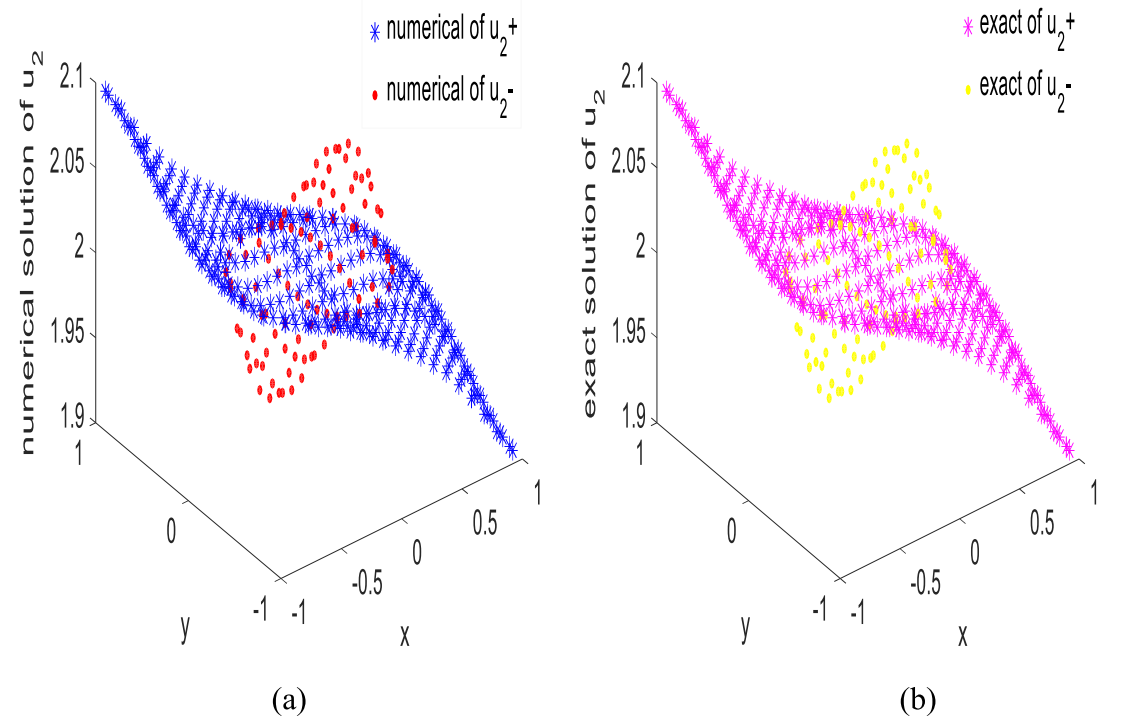


Fig. 4. The (a) numerical and (b) exact solutions of u_2 with $N_x = 20$ for Example 1.

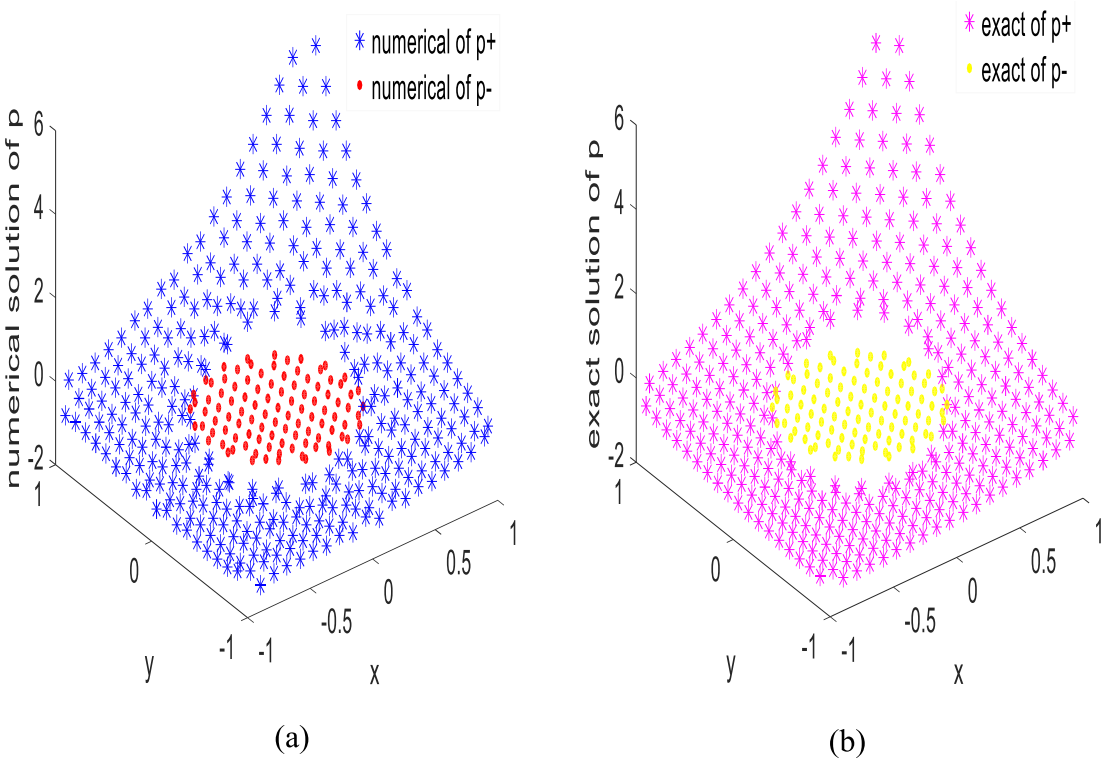


Fig. 5. The (a) numerical and (b) exact solutions of p with $N_x = 20$ for Example 1.

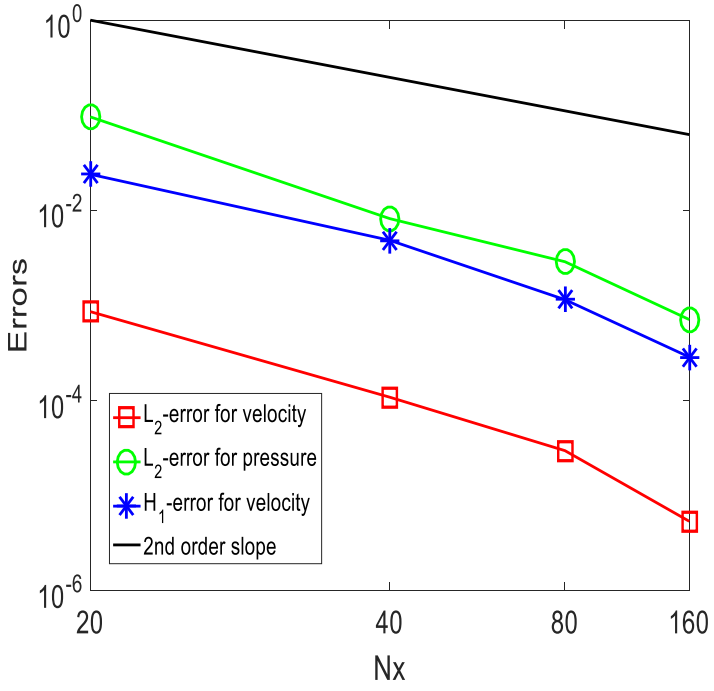


Fig. 6. The L_2, H_1 error on log-log scale with different N_x in $m = 16$ for Example 1.

$d_k (k = 1, 2, ..., m)$. Obviously, the weighting function ω_k is used to indicate the importance of u_k to u_0.

Denote the partial derivatives of $u(x)$ at node (x_0,y_0) appeared in $B(u)$ by

$$D_u = \left\{ \frac{\partial u_0}{\partial x}, \frac{\partial u_0}{\partial y}, \frac{\partial^2 u_0}{\partial x^2}, \frac{\partial^2 u_0}{\partial y^2}, \frac{\partial^2 u_0}{\partial x \partial y} \right\}^T. \tag{20}$$

And their coefficients are given by

$$q_k = \left\{ h_k, l_k, \frac{h_k^2}{2}, \frac{l_k^2}{2}, h_k l_k \right\}^T, \quad (k = 1, 2, ..., m). \tag{21}$$

We define $Q=[q_1,q_2,\cdots, q_m]^T$. The residual function $B(u)$ can be rewritten as a matrix formula as follows:

Table 1
The L_2 and H_1 error of u_1 with $m = 16$ for Example 1.

N_x	GFDM for MBC				IFEM [6]			
	e_{u_1}	rate	$e_{u_1,D}$	rate	e_{u_1}	rate	$e_{u_1,D}$	rate
40	7.6076e-05		3.4143-03		4.3181e-04		2.3170e-02	
80	2.0849e-05	1.87	8.1072e-04	2.07	1.1374e-04	1.92	1.1938e- 02	0.96
160	3.7719e-06	2.47	2.0188e-04	2.00	2.9448e-05	1.95	6.1789e-03	0.95

Table 2
The L_2 and H_1 error of u_2 with $m = 16$ for Example 1.

N_x	GFDM for MBC				IFEM [6]			
	e_{u_2}	rate	$e_{u_2,D}$	rate	e_{u_2}	rate	$e_{u_2,D}$	rate
40	7.6528e-05		3.4135e-03		4.3180e-04		2.3170e-02	
80	2.0942e-05	1.87	8.1069e-04	2.07	1.1374e-04	1.92	1.1938e-02	0.96
160	3.7804e-06	2.47	2.0186e-04	2.01	2.9448e-05	1.95	6.1789e- 03	0.95

$$B(u) = (QD_u + \overrightarrow{u_0} - \overrightarrow{u})^T W(QD_u + \overrightarrow{u_0} - \overrightarrow{u}), \tag{22}$$

where $\overrightarrow{u} = (u_1, u_2, \cdots, u_m)^T$, $\overrightarrow{u_0} = (u_0, u_0, \cdots, u_0)^T$and $W = diag(\omega_1^2, \omega_2^2, \cdots, \omega_m^2)$.

To minimize the above residual function $B(u)$ with respect to the partial derivatives D_u, the linear equation system can be obtained,

$$AD_u = b, \tag{23}$$

where A is a symmetric matrix given by

$$A = Q^T WQ. \tag{24}$$

And the right hand is re-written as follows:

$$b = Q^T W(\overrightarrow{u} - \overrightarrow{u_0}) = \left(-\sum_{i=1}^{m} \omega_i^2 q_i^T, \omega_1^2 q_1^T, \cdots, \omega_m^2 q_m^T \right) U = BU, \tag{25}$$

where $U = (u_0, u_1, \cdots, u_m)^T$are the functional values at the node (x_0, y_0) and the other nodes (x_k, y_k) inside the star. In addition, the size of the matrix A is 5×5 in the 2D case. Then,

$$D_u = A^{-1}b = A^{-1}(BU) = \left(A^{-1}B\right)U = EU, \tag{26}$$

where

$$E = A^{-1}B. \tag{27}$$

In addition, the derivative of the central node (x_0, y_0) can be approximated by the linear combination of the function values of itself and its m neighboring nodes, as shown in Eq. (26). For example,

$$\frac{\partial u_0}{\partial x} = \sum_{j=1}^{m} E_{1j} u_j^0, \quad \frac{\partial^2 u_0}{\partial x^2} = \sum_{j=1}^{m} E_{3j} u_j^0. \tag{28}$$

It means that the first-order derivative and second-order derivative of the function value at the central node (x_0, y_0) can be written as the procedure of the matrix formed by the function values of nodes inside the star and the first row and the third row of the matrix E formed by Eq. (27), respectively.

Furthermore, by enforcing the satisfactions of the governing equations at the interior nodes and enforcing boundary conditions at the boundary nodes, a system of linear algebraic equations can be created. Finally, the numerical results can be obtained by solving the linear matrix system.

4. GFDM for Stokes interface problem with a mixed boundary condition

We use the GFDM to study the Stokes interface problem (7)-(13). Firstly, we make N_{in}^+, N_{in}^-, N_b^+, N_b^-, N_Γ^-, N_Γ^+ as numbers of scattered nodes in the interior of the domain Ω^+, Ω^-, on the domain boundary $\partial\Omega^+$, $\partial\Omega^-$, and on the interface Γ^-,Γ^+, respectively. Next, we decompose the Stokes interface problem into two sub-problems, which are coupled by two interface conditions. It can be written as follows:

$$-\mu^-(x_i)\Delta \boldsymbol{u}^-(x_i) + \nabla p^-(x_i) = \boldsymbol{f}^-(x_i), x_i \in \Omega^-, \tag{29}$$

$$\nabla \cdot \boldsymbol{u}^-(x_i) = 0, x_i \in \Omega^-, \tag{30}$$

$$\boldsymbol{u}^-(x_i) = \boldsymbol{g}^-(x_i) = \left(g_1^-(x_i), g_2^-(x_i)\right), x_i \in \partial\Omega^-, \tag{31}$$

$$-\mu^-(x_i)\Delta \boldsymbol{u}^-(x_i)\cdot \boldsymbol{n_b} + \nabla p^-(x_i)\cdot \boldsymbol{n_b} = \boldsymbol{f}^-(x_i)\cdot \boldsymbol{n_b}, x_i \in \partial\Omega^-, \tag{32}$$

$$[\boldsymbol{u}(x_i)]_\Gamma = \boldsymbol{u}^+(x_i) - \boldsymbol{u}^-(x_i) = \boldsymbol{\varphi}(x_i), x_i \in \Gamma^-, \tag{33}$$

$$-\mu^-(x_i)\Delta \boldsymbol{u}^-(x_i)\cdot \boldsymbol{n}^- + \nabla p^-(x_i)\cdot \boldsymbol{n}^- = \boldsymbol{f}^-(x_i)\cdot \boldsymbol{n}^-, x_i \in \Gamma^-, \tag{34}$$

and

$$-\mu^+(x_i)\Delta \boldsymbol{u}^+(x_i) + \nabla p^+(x_i) = \boldsymbol{f}^+(x_i), x_i \in \Omega^+, \tag{35}$$

$$\nabla \cdot \boldsymbol{u}^+(x_i) = 0, x_i \in \Omega^+, \tag{36}$$

$$\boldsymbol{u}^+(x_i) = \boldsymbol{g}^+(x_i) = \left(g_1^+(x_i), g_2^+(x_i)\right), x_i \in \partial\Omega^+, \tag{37}$$

$$-\mu^+(x_i)\Delta \boldsymbol{u}^+(x_i)\cdot \boldsymbol{n_b} + \nabla p^+(x_i)\cdot \boldsymbol{n_b} = \boldsymbol{f}^+(x_i)\cdot \boldsymbol{n_b}, x_i \in \partial\Omega^+, \tag{38}$$

$$\begin{aligned}[\sigma(\boldsymbol{u}, p)\cdot \boldsymbol{n}]_\Gamma &= \sigma(\boldsymbol{u}^+, p^+)\cdot \boldsymbol{n}^+ + \sigma(\boldsymbol{u}^-, p^-)\cdot \boldsymbol{n}^- \\ &= \boldsymbol{\psi}(x_i), x_i \in \Gamma^+,\end{aligned} \tag{39}$$

$$-\mu^+(x_i)\Delta \boldsymbol{u}^+(x_i)\cdot \boldsymbol{n}^+ + \nabla p^+(x_i)\cdot \boldsymbol{n}^+ = \boldsymbol{f}^+(x_i)\cdot \boldsymbol{n}^+, x_i \in \Gamma^+, \tag{40}$$

where $\boldsymbol{n}^- = (n_x^-, n_y^-)$, $\boldsymbol{n}^+ = (n_x^+, n_y^+)$ are the normal vectors of the interface Γ^-, Γ^+, respectively.

We note that the interface condition on Γ serves as the boundary condition of the sub-problems. Obviously, the geometry of the interface only affects the distribution of points on the boundary of the sub-problems. In other words, the formulation (29)-(40) has advantages to deal with the interface with a complex geometric shape.

We make the N_b^-, N_b^+ boundary points, N_{in}^-, N_{in}^+ interior points and N_Γ^-, N_Γ^+ interface points satisfy the boundary conditions, governing equa-

Table 3
The L_2 error of p with $m = 16$ for Example 1.

N_x	GFDM for MBC		IFEM [6]		GFDM for PPE	
	e_p	rate	e_p	rate	e_p	rate
40	8.165e-03		7.092e-02		1.245e-02	
80	2.872e-03	1.51	3.540e-02	1.00	3.664e-03	1.76
160	7.077e-04	2.02	1.772e-02	0.99	1.586e-03	1.21

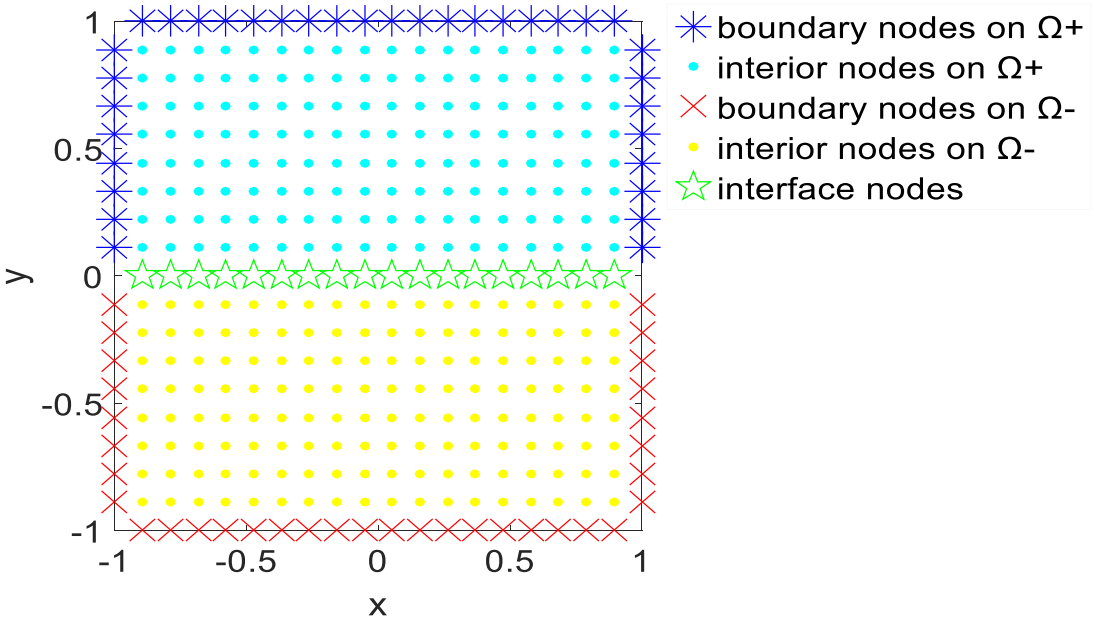


Fig. 7. The point collocation with $N_x = 20$ for Example 2.

tions and interface conditions, respectively. We define the number of the total nodes as follows:

$$N = N_{in}^{+} + N_{in}^{-} + N_b^{+} + N_b^{-} + N_\Gamma^{+} + N_\Gamma^{-}. \tag{41}$$

By combining above $3N$ linear algebraic equations, we write the final equation in the following matrix form:

$$\begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \begin{pmatrix} U_1 \\ U_2 \\ P \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \tag{42}$$

where

$$U_j = \left(U_j^{-}, U_j^{+}\right)^T, (j=1,2), \tag{43}$$

$$P = (P^{-}, P^{+})^T, \tag{44}$$

$$b_j = \left(b_j^{-}, b_j^{+}\right)^T, (j=1,2,3). \tag{45}$$

Where

$$\begin{aligned} U_j^{-} &= \left(u_{j,1}^{-}, \cdots, u_{j,N_\Gamma^{-}}^{-}, u_{j,N_\Gamma^{-}+1}^{-}, \cdots, u_{j,N_\Gamma^{-}+N_b^{-}}^{-}, \right. \\ &\quad \left. u_{j,N_\Gamma^{-}+N_b^{-}+1}^{-}, \cdots, u_{j,N_\Gamma^{-}+N_b^{-}+N_{in}^{-}}^{-}\right)^T, (j=1,2), \end{aligned} \tag{46}$$

$$\begin{aligned} U_j^{+} &= \left(u_{j,1}^{+}, \cdots, u_{j,N_\Gamma^{+}}^{+}, u_{j,N_\Gamma^{+}+1}^{+}, \cdots, u_{j,N_\Gamma^{+}+N_b^{+}}^{+}, \right. \\ &\quad \left. u_{j,N_\Gamma^{+}+N_b^{+}+1}^{+}, \cdots, u_{j,N_\Gamma^{+}+N_b^{+}+N_{in}^{+}}^{+}\right)^T, (j=1,2), \end{aligned} \tag{47}$$

$$\begin{aligned} P^{-} &= \left(p_1^{-}, \cdots, p_{N_\Gamma^{-}}^{-}, p_{N_\Gamma^{-}+1}^{-}, \cdots, p_{N_\Gamma^{-}+N_b^{-}}^{-}, p_{N_\Gamma^{-}+N_b^{-}+1}^{-}, \right. \\ &\quad \left. \cdots, p_{N_\Gamma^{-}+N_b^{-}+N_{in}^{-}}^{-}\right)^T \end{aligned} \tag{48}$$

$$\begin{aligned} P^{+} &= \left(p_1^{+}, \cdots, p_{N_\Gamma^{+}}^{+}, p_{N_\Gamma^{+}+1}^{+}, \cdots, p_{N_\Gamma^{+}+N_b^{+}}^{+}, p_{N_\Gamma^{+}+N_b^{+}+1}^{+}, \right. \\ &\quad \left. \cdots, p_{N_\Gamma^{+}+N_b^{+}+N_{in}^{+}}^{+}\right)^T, \end{aligned} \tag{49}$$

$$\begin{aligned} b_j^{-} &= \left(\varphi_j\left(x_1^{-}\right), \cdots, \varphi_j\left(x_{N_\Gamma^{-}}^{-}\right), g_j^{-}\left(x_{N_\Gamma^{-}+1}^{-}\right), \cdots g_j^{-}\left(x_{N_\Gamma^{-}+N_b^{-}}^{-}\right), \right. \\ &\quad \left. f_j^{-}\left(x_{N_\Gamma^{-}+N_b^{-}+1}^{-}\right), \cdots, f_j^{-}\left(x_{N_\Gamma^{-}+N_b^{-}+N_{in}^{-}}^{-}\right)\right)^T, (j=1,2), \end{aligned} \tag{50}$$

$$\begin{aligned} b_j^{+} &= \left(\psi_j(x_1^{+}), \cdots, \psi_j\left(x_{N_\Gamma^{+}}^{+}\right), g_j^{+}\left(x_{N_\Gamma^{+}+1}^{+}\right), \cdots, g_j^{+}\left(x_{N_\Gamma^{+}+N_b^{+}}^{+}\right), \right. \\ &\quad \left. f_j^{+}\left(x_{N_\Gamma^{+}+N_b^{+}+1}^{+}\right), \cdots, f_j^{+}\left(x_{N_\Gamma^{+}+N_b^{+}+N_{in}^{+}}^{+}\right)\right)^T, (j=1,2), \end{aligned} \tag{51}$$

$$\begin{aligned} b_3^{-} &= \left(\left(f_1^{-}\cdot n_{bx} + f_2^{-}\cdot n_{by}\right)\left(x_1^{-}\right), \cdots, \left(f_1^{-}\cdot n_{bx} + f_2^{-}\cdot n_{by}\right)\left(x_{N_\Gamma^{-}}^{-}\right), \right. \\ &\quad \left(f_1^{-}\cdot n_{bx} + f_2^{-}\cdot n_{by}\right)\left(x_{N_\Gamma^{-}+1}^{-}\right), \cdots, \left(f_1^{-}\cdot n_{bx} + f_2^{-}\cdot n_{by}\right)\left(x_{N_\Gamma^{-}+N_b^{-}}^{-}\right), \\ &\quad \left. 0, \cdots, 0\right)^T, \end{aligned} \tag{52}$$

$$\begin{aligned} b_3^{+} &= \left(\left(f_1^{+}\cdot n_{bx} + f_2^{+}\cdot n_{by}\right)\left(x_1^{+}\right), \cdots, \left(f_1^{+}\cdot n_{bx} + f_2^{+}\cdot n_{by}\right)\left(x_{N_\Gamma^{+}}^{+}\right), \right. \\ &\quad \left(f_1^{+}\cdot n_{bx} + f_2^{+}\cdot n_{by}\right)\left(x_{N_\Gamma^{+}+1}^{+}\right), \cdots, \left(f_1^{+}\cdot n_{bx} + f_2^{+}\cdot n_{by}\right)\left(x_{N_\Gamma^{+}+N_b^{+}}^{+}\right), \\ &\quad \left. 0, \cdots, 0\right)^T. \end{aligned} \tag{53}$$

And

$$A_1 = B_2 = \begin{bmatrix} E_{N_\Gamma^{-}\times N_\Gamma^{-}} & 0 & 0 & -E_{N_\Gamma^{+}\times N_\Gamma^{+}} & 0 & 0 \\ 0 & E_{N_b^{-}\times N_b^{-}} & 0 & 0 & 0 & 0 \\ 0 & 0 & A_{N_{in}^{-}\times N_{in}^{-}}^{-} & 0 & 0 & 0 \\ A_{N_\Gamma^{-}\times N_\Gamma^{-}}^{\Gamma^{-}} & 0 & 0 & A_{N_\Gamma^{+}\times N_\Gamma^{+}}^{\Gamma^{+}} & 0 & 0 \\ 0 & 0 & 0 & 0 & E_{N_b^{+}\times N_b^{+}} & 0 \\ 0 & 0 & 0 & 0 & 0 & A_{N_{in}^{+}\times N_{in}^{+}}^{+} \end{bmatrix}, \tag{54}$$

$$A_2 = B_1 = [0]_{N\times N}, \tag{55}$$

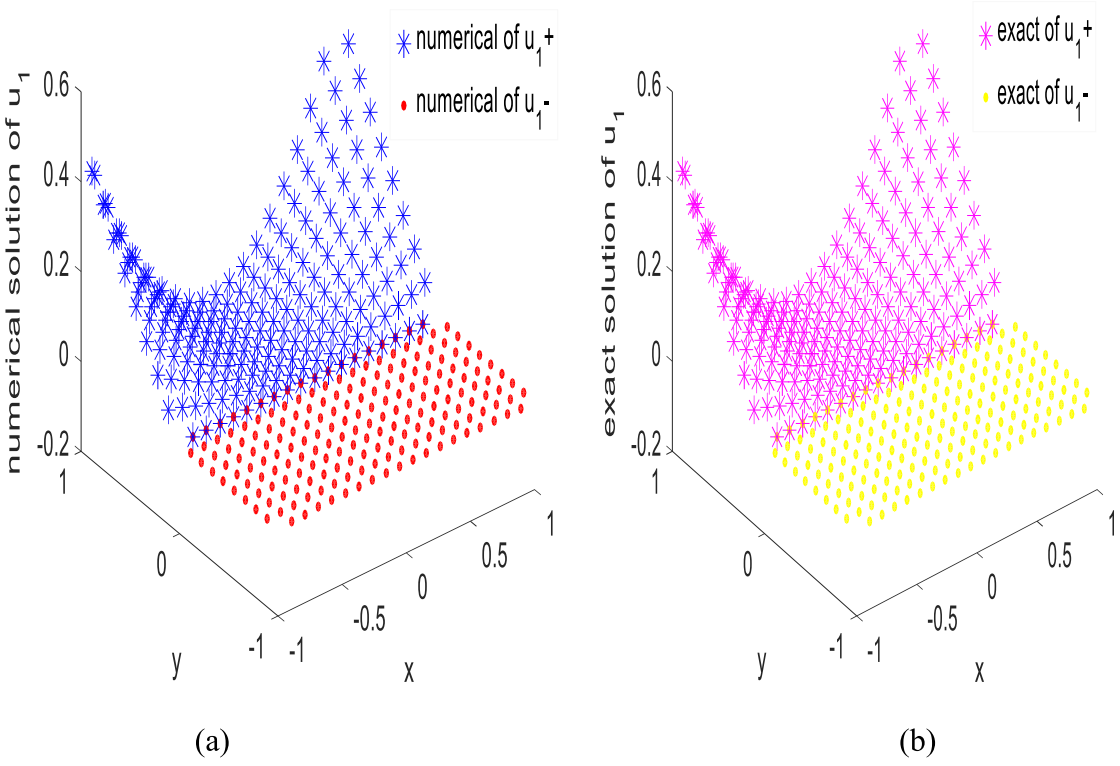


Fig. 8. The (a) numerical and (b) exact solutions of u_1 with $N_x = 20$ for Example 2.

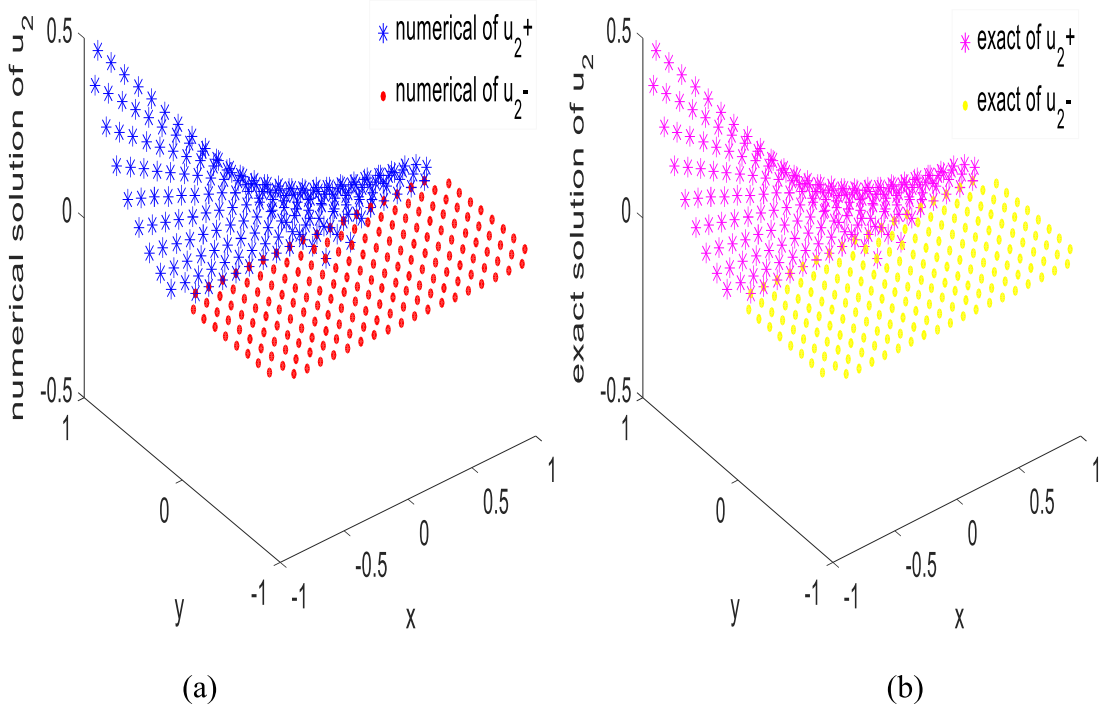


Fig. 9. The (a) numerical and (b) exact solutions of u_2 with $N_x = 20$ for Example 2.

$$A_3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & B^{-}_{N^{-}_{in}\times N^{-}_{in}} & 0 & 0 & 0 \\ B^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}} & 0 & 0 & B^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & B^{+}_{N^{+}_{in}\times N^{+}_{in}} \end{bmatrix}, \tag{56}$$

$$B_3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & C^{-}_{N^{-}_{in}\times N^{-}_{in}} & 0 & 0 & 0 \\ C^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}} & 0 & 0 & C^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & C^{+}_{N^{+}_{in}\times N^{+}_{in}} \end{bmatrix}, \tag{57}$$

$$\begin{aligned} C_1 = \mathrm{diag}(&D^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}, D^{-}_{N^{-}_{b}\times N^{-}_{b}}, D^{-}_{N^{-}_{in}\times N^{-}_{in}}, \\ &D^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}, D^{+}_{N^{+}_{b}\times N^{+}_{b}}, D^{+}_{N^{+}_{in}\times N^{+}_{in}}), \end{aligned} \tag{58}$$

$$\begin{aligned} C_2 = \mathrm{diag}(&F^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}, F^{-}_{N^{-}_{b}\times N^{-}_{b}}, F^{-}_{N^{-}_{in}\times N^{-}_{in}}, \\ &F^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}, F^{+}_{N^{+}_{b}\times N^{+}_{b}}, F^{+}_{N^{+}_{in}\times N^{+}_{in}}), \end{aligned} \tag{59}$$

$$\begin{aligned} C_3 = \mathrm{diag}(&G^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}, G^{-}_{N^{-}_{b}\times N^{-}_{b}}, 0_{N^{-}_{in}\times N^{-}_{in}}, \\ &G^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}, G^{+}_{N^{+}_{b}\times N^{+}_{b}}, 0_{N^{+}_{in}\times N^{+}_{in}}), \end{aligned} \tag{60}$$

where

$$\begin{cases} A^{-}_{N^{-}_{in}\times N^{-}_{in}}(i,:) = -\mu_1(E(3,i)+E(4,i)), \\ A^{+}_{N^{+}_{in}\times N^{+}_{in}}(i,:) = -\mu_2(E(3,i)+E(4,i)), \\ B^{-}_{N^{-}_{in}\times N^{-}_{in}}(i,:) = B^{+}_{N^{+}_{in}\times N^{+}_{in}}(i,:) = E(1,i), \\ D^{-}_{N^{-}_{in}\times N^{-}_{in}}(i,:) = D^{+}_{N^{+}_{in}\times N^{+}_{in}}(i,:) = E(1,i), \\ C^{-}_{N^{-}_{in}\times N^{-}_{in}}(i,:) = C^{+}_{N^{+}_{in}\times N^{+}_{in}}(i,:) = E(2,i), \\ F^{-}_{N^{-}_{in}\times N^{-}_{in}}(i,:) = F^{+}_{N^{+}_{in}\times N^{+}_{in}}(i,:) = E(2,i), \end{cases} \tag{61}$$

$$\begin{cases} A^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}(i,:) = \mu_1\left(E(1,i)n^{-}_{x} + E(2,i)n^{-}_{y}\right), \\ B^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}(i,:) = -n^{-}_{x}, \\ C^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}(i,:) = -n^{-}_{y}, \\ D^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}(i,:) = -\mu_1(E(3,i)+E(4,i))n^{-}_{x}, \\ F^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}(i,:) = -\mu_1(E(3,i)+E(4,i))n^{-}_{y}, \\ G^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}(i,:) = E(1,i)n^{-}_{x} + E(2,i)n^{-}_{y}, \end{cases} \tag{62}$$

$$\begin{cases} A^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}(i,:) = \mu_2\left(E(1,i)n^{+}_{x} + E(2,i)n^{+}_{y}\right), \\ B^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}(i,:) = -n^{+}_{x}, \\ C^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}(i,:) = -n^{+}_{y}, \\ D^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}(i,:) = -\mu_2(E(3,i)+E(4,i))n^{+}_{x}, \\ F^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}(i,:) = -\mu_2(E(3,i)+E(4,i))n^{+}_{y}, \\ G^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}(i,:) = E(1,i)n^{+}_{x} + E(2,i)n^{+}_{y}, \end{cases} \tag{63}$$

$$\begin{cases} D^{-}_{N^{-}_{b}\times N^{-}_{b}}(i,:) = -\mu_1(E(3,i)+E(4,i))n_{bx}, \\ D^{+}_{N^{+}_{b}\times N^{+}_{b}}(i,:) = -\mu_2(E(3,i)+E(4,i))n_{bx}, \\ F^{-}_{N^{-}_{b}\times N^{-}_{b}}(i,:) = -\mu_1(E(3,i)+E(4,i))n_{by}, \\ F^{+}_{N^{+}_{b}\times N^{+}_{b}}(i,:) = -\mu_2(E(3,i)+E(4,i))n_{by}, \\ G^{-}_{N^{-}_{b}\times N^{-}_{b}}(i,:) = G^{+}_{N^{+}_{b}\times N^{+}_{b}}(i,:) = E(1,i)n_{bx} + E(2,i)n_{by}. \end{cases} \tag{64}$$

Where the matrix $E(j, i)(j = 1, \cdots 4)$ is the j-th row and the i-th column of the matrix E formed by Eq. (27).

The governing Eqs. (29) and (35) lead to the matrix$A^{-}_{N^{-}_{in}\times N^{-}_{in}}$,$B^{-}_{N^{-}_{in}\times N^{-}_{in}}$,$C^{-}_{N^{-}_{in}\times N^{-}_{in}}$ and the matrix $A^{+}_{N^{+}_{in}\times N^{+}_{in}}$,$B^{+}_{N^{+}_{in}\times N^{+}_{in}}$,$C^{+}_{N^{+}_{in}\times N^{+}_{in}}$ in (54)-(57), respectively. The governing Eqs. (30) and (36) lead to the matrix $D^{-}_{N^{-}_{in}\times N^{-}_{in}}$,$F^{-}_{N^{-}_{in}\times N^{-}_{in}}$ and matrix $D^{+}_{N^{+}_{in}\times N^{+}_{in}}$,$F^{+}_{N^{+}_{in}\times N^{+}_{in}}$ in (58)-(59), respectively.

The Dirichlet boundary condition (31) and (37) leads the unit matrix $E_{N^{-}_{b}\times N^{-}_{b}}$ and $E_{N^{+}_{b}\times N^{+}_{b}}$ in (57). The mixed boundary conditions (32), (38) and interface conditions (34), (40) lead to the matrix $D^{-}_{N^{-}_{b}\times N^{-}_{b}}$, $F^{-}_{N^{-}_{b}\times N^{-}_{b}}$, $G^{-}_{N^{-}_{b}\times N^{-}_{b}}$, $D^{+}_{N^{+}_{b}\times N^{+}_{b}}$, $F^{+}_{N^{+}_{b}\times N^{+}_{b}}$, $G^{+}_{N^{+}_{b}\times N^{+}_{b}}$ and $D^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $F^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $G^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $D^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$, $F^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$, $G^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$ in (58)-(60), respectively. The interface conditions (33) and (39) lead to the unit matrix $E_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $E_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$ in (54) and matrix $A^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $A^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$ $B^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $B^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$ $C^{\Gamma^{-}}_{N^{-}_{\Gamma}\times N^{-}_{\Gamma}}$, $C^{\Gamma^{+}}_{N^{+}_{\Gamma}\times N^{+}_{\Gamma}}$ in (54)-(57), respectively. These show that GFDM has notable advantages to deal with the complex boundary conditions, such as Eqs. (10)–(13) can be easily discretized by the GFDM. Because the derivative of unknown variables is approximate by a linear combination of the function values on neighboring nodes in GFDM.

5. Numerical results

In this section, three numerical examples are proposed to verify the accuracy and stability of the GFDM for solving the Stokes interface problem. In Example 1, we consider the problem with a circular interface and compare the accuracy of the GFDM with the discontinuous IFEM in [6]. In Example 2, we consider a linear interface and compare the numerical results of GFDM to solve the MBC form (7)-(13) and the classical PPE form (14) of the Stokes interface problem. In Example 3, we consider the stokes interface problem with a hexagon interface and give numerical results. Furthermore, we test the influences of the parameter m and the discontinuity of the viscosity coefficient μ in Section 5.4.

The following symbols are used in these numerical examples: $N_x = N_y$ and N_x, N_y are the number of scattered nodes on the x-axis and y-axis, respectively. We define the L_2 norm error of the velocity $\boldsymbol{u} = (u_1,u_2)$ and the fluid pressure field p as follows:

$$e_{u_j} = \left[\sum_{i=1}^{N}\frac{(u_j^i - u_j(x_i))^2}{N}\right]^{\frac{1}{2}} (j = 1, 2), \quad e_u = [e^2_{u_1} + e^2_{u_2}]^{\frac{1}{2}}, \quad e_p = \left[\sum_{i=1}^{N}\frac{(p_i - p(x_i))^2}{N}\right]^{\frac{1}{2}}.$$

And, the H_1 norm error related to the velocity is defined as follows:

$$e_{u_j,D} = \left[\sum_{i=1}^{N}\frac{(\nabla u_j^i - \nabla u_j(x_i))^2}{N}\right]^{\frac{1}{2}} (j = 1,2), e_{u,D} = [e^2_{u_1,D} + e^2_{u_2,D}]^{\frac{1}{2}}, \text{where } (u_1^i, u_2^i, p_i) \text{ and } (u_1(x_i), u_2(x_i), p(x_i)) \text{ are numerical and exact solutions at the node } x_i = (x_1^i, x_2^i), \text{ respectively.}$$

5.1. Example 1: the Stokes interface problem with a circular interface

We consider a Stokes interface problem given in [6] with the domain $\Omega = [-1, 1] \times [-1, 1]$ cut by a circular interface Γ: $\{(x, y)|x^2 + y^2 = 0.3\}$. We will compare with the numerical results of the sample

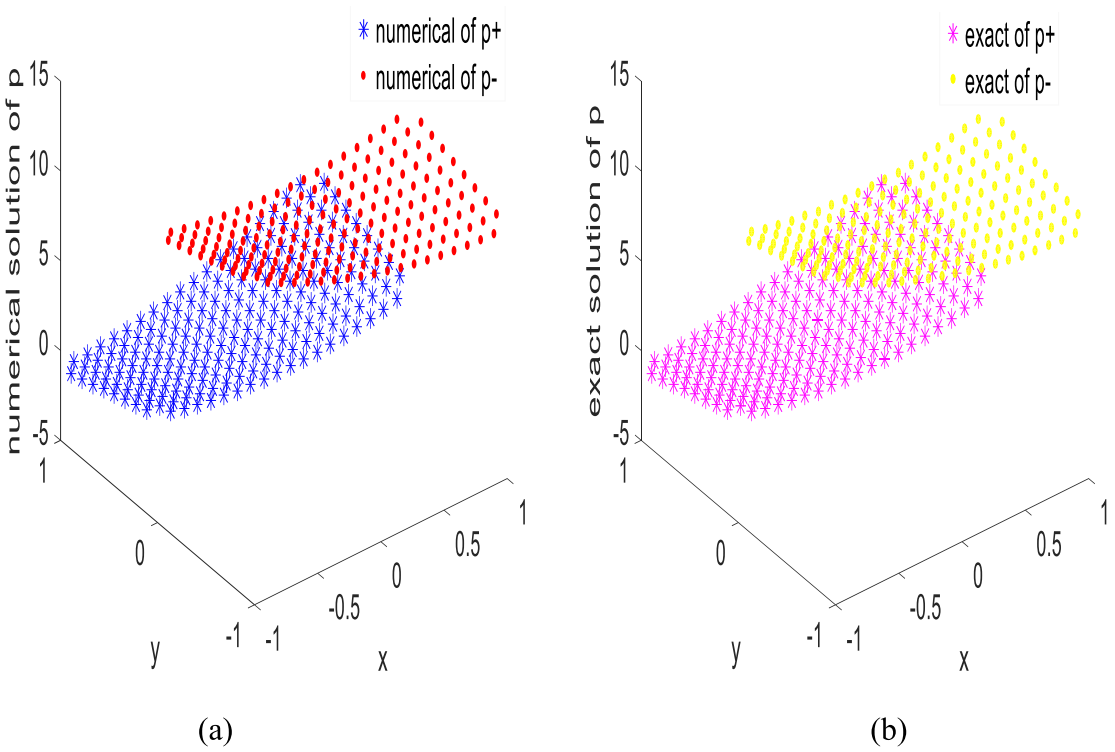


Fig. 10. The (a) numerical and (b) exact solutions of p with $N_x = 20$ for Example 2.

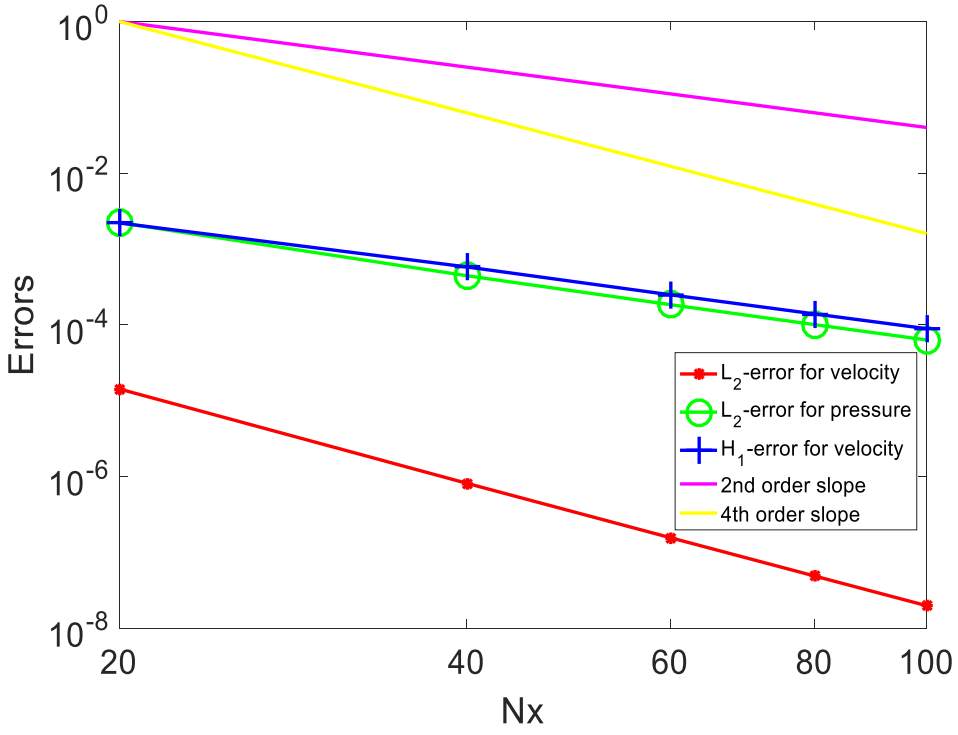


Fig. 11. The L_2,H_1 error on the log-log scale with different N_x in $m = 16$ for Example 2.

Table 4
The L_2 and H_1 error of velocity with $m = 16$ for Example 2.

	GFDM for MBC				GFDM for PPE			
N_x	e_u	rate	$e_{u,D}$	rate	e_u	rate	$e_{u,D}$	rate
20	1.4282e-05		2.2268e-03		1.4798e-04		3.9303e-03	
40	8.1327e-07	4.13	4.4386e-04	2.33	7.9939e-06	4.21	5.7891e-04	2.76
60	1.5606e-07	4.07	1.8445e-04	2.17	1.5089e-06	4.11	2.1339e-04	2.46
80	4.8696e-08	4.05	1.0054e-04	2.11	4.6699e-07	4.08	1.1003e-04	2.30

questions given in Reference [6]. Let $\Omega^+ = \{(x, y)|x^2 + y^2 > 0.3\}$ and $\Omega^- = \{(x, y)|x^2 + y^2 < 0.3\}$. And the piecewise constant viscosity coefficient is defined as follows:

$$\mu = \begin{cases} 10, (x, y) \in \Omega^+, \\ 1, (x, y) \in \Omega^-. \end{cases}$$

We define the right hand by $\boldsymbol{f} = -\mu\Delta\boldsymbol{u} + \nabla p$, the boundary data $\boldsymbol{g}$ and the interface condition ϕ and ψ with the following exact solutions $\boldsymbol{u} = (u_1, u_2)$ and p:

$$u_1 = \begin{cases} 1 - \frac{y}{\mu^+}\sin\left(0.3 - x^2 - y^2\right), (x, y) \in \Omega^+, \\ 1 - \frac{y}{\mu^-}\sin\left(0.3 - x^2 - y^2\right), (x, y) \in \Omega^-, \end{cases}$$

$$u_2 = \begin{cases} 2 + \frac{x}{\mu^+}\sin\left(0.3 - x^2 - y^2\right), (x, y) \in \Omega^+, \\ 2 + \frac{x}{\mu^-}\sin\left(0.3 - x^2 - y^2\right), (x, y) \in \Omega^-, \end{cases}$$

Table 5
The L_2 error of the pressure with $m = 16$ for Example 2.

N_x	GFDM for MBC		GFDM for PPE	
	e_p	rate	e_p	rate
20	2.2067e-03		2.7871e-01	
40	5.7943e-04	1.93	6.0355e-02	2.21
60	2.4904e-04	2.08	2.5698e-02	2.11
80	1.3916e-04	2.02	1.4161e-02	2.07

$$p = \begin{cases} p^+ = e^{x+y} - 1.3798535909816816, (x, y) \in \Omega^+, \\ p^- = \sqrt{1 + x^2 + y^2} - 1.3798535909816816, (x, y) \in \Omega^-. \end{cases}$$

The interior nodes and boundary nodes are evenly distributed within the computational and along the boundary (see Fig. 2). The numerical and exact solutions of u_1, u_2 and p by using the MBC form are shown in Figs. 3–5, respectively. Those figures show that the numerical solutions are compared well with the exact solutions.

The log-log plot of L_2 errors of the velocity and the pressure, and H_1 errors of the velocity with different N_x in $m = 16$ are shown in Fig. 6. In Fig. 6, the convergence rates of the error curves are stable. Meanwhile, the L_2 error of the velocity and pressure, and the H_1 error of the velocity maintain the 2nd-order convergence.

Furthermore, the comparison of the numerical results of the GFDM and the discontinuous IFEM method are shown in Tables 1–3. We can know that the two methods yield accurate numerical results, but the errors of the GFDM for the MBC form are much less than the IFEM. It is worth noticing that the errors $e_{u_1,D}, e_{u_2,D}$ and e_p of GFDM for the MBC form can reach 2nd-order convergence, while $e_{u_1,D}, e_{u_2,D}$ and e_p of the IFEM only obtain 1st-order convergence. In addition, we also compare the L_2 errors of the pressure between the GFDM for the MBC form and the PPE form in Table 3. It shows that the MBC form is more accurate than the PPE form for this example.

5.2. Example 2: the Stokes interface problem with a linear interface

In the second example, we consider a numerical example [5] with the computational domain $\Omega = [-1, 1] \times [-1, 1]$ and a linear interface Γ: $\{(x, y)|y = 0\}$. In this problem, the piecewise constant viscosity

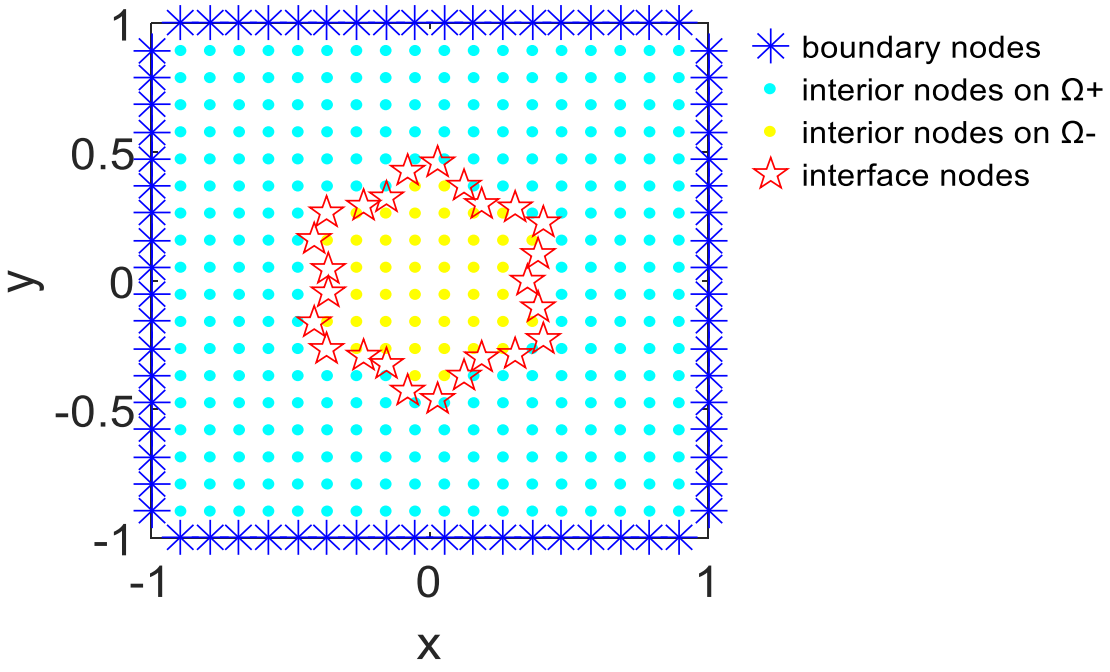


Fig. 12. The point collocation with $N_x = 20$ for Example 3.

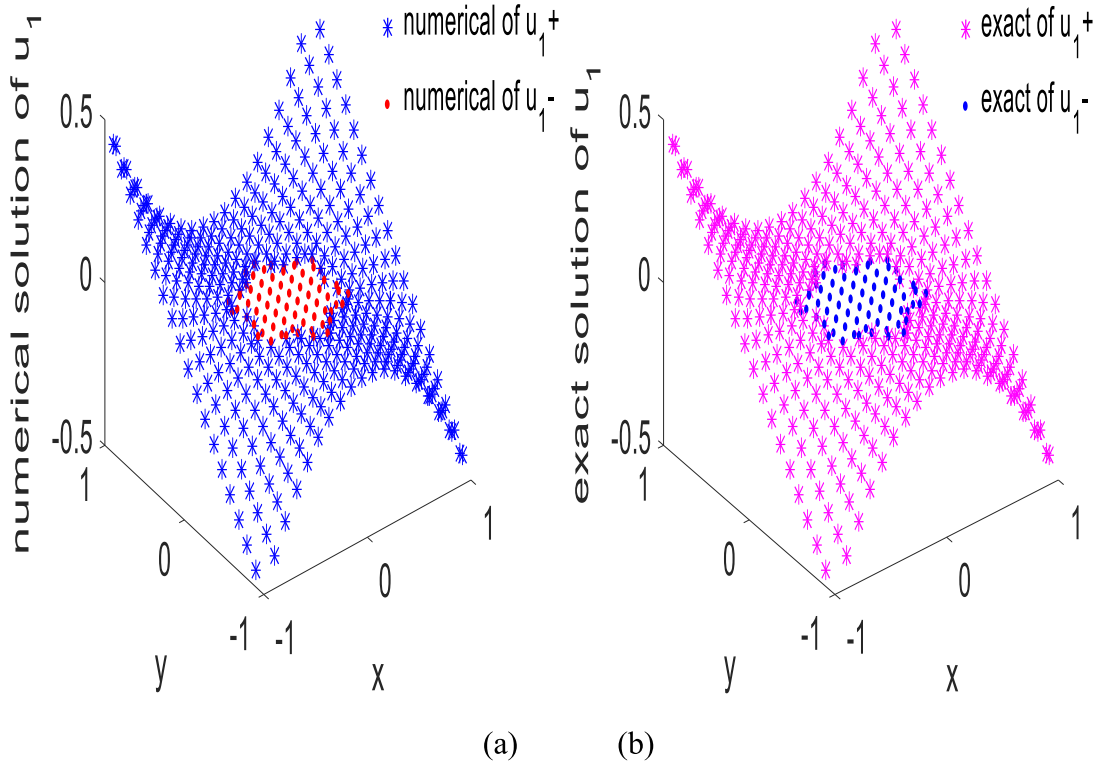


Fig. 13. The (a) numerical and (b) exact solutions of u_1 with $N_x = 20$ for Example 3.

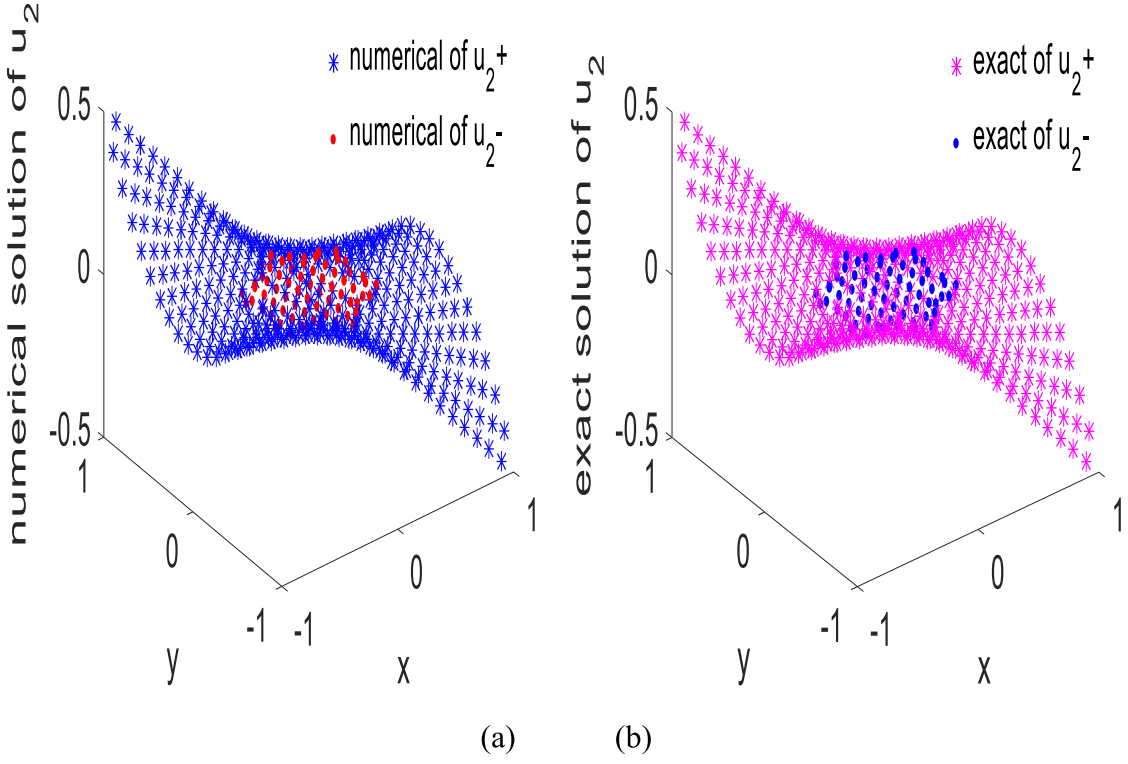


Fig. 14. The (a) numerical and (b) exact solutions of u_2 with $N_x = 20$ for Example 3.

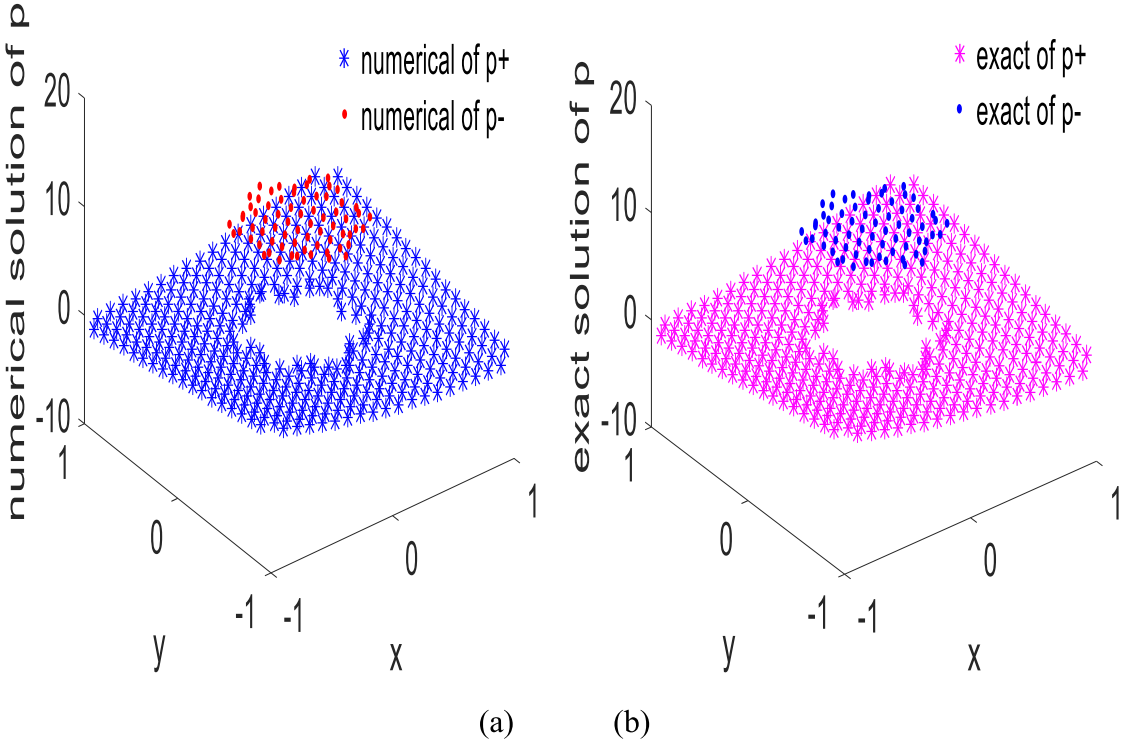


Fig. 15. The (a) numerical and (b) exact solutions of p with $N_x = 20$ for Example 3.

Table 6
The errors of velocity and pressure with $m = 16$ for Example 3.

N_x	The GFDM for MBC					
	e_u	rate	$e_{u,D}$	rate	e_p	rate
40	1.8334e-04		8.8968e-04		2.7191e-02	
70	5.0873e-06	6.4054	1.4784e-04	3.2071	2.5448e-03	4.233
100	5.9031e-07	6.0387	6.4598e-05	2.3213	5.8935e-04	4.1011

coefficient is defined as follows:

$$\mu = \begin{cases} 2, (x,y) \in \Omega^+, \\ 200, (x,y) \in \Omega^-, \end{cases}$$

where $\Omega^+ = \{(x,y) | y > 0\}$ and $\Omega^- = \{(x, y) | y < 0\}$.

We define the Dirichlet boundary data $\boldsymbol{g}$ for the velocity with the following exact solutions $\boldsymbol{u} = (u_1, u_2)$ and p:

$$u_1 = \begin{cases} \frac{1}{\mu^+} x^2 y, (x,y) \in \Omega^+, \\ \frac{1}{\mu^-} x^2 y, (x,y) \in \Omega^-, \end{cases} \quad u_2 = \begin{cases} -\frac{1}{\mu^+} xy^2, (x,y) \in \Omega^+, \\ -\frac{1}{\mu^-} xy^2, (x,y) \in \Omega^-, \end{cases}$$

$$p = 2xy + x^2 + 10\chi(y),$$

where

$$\chi(y) = \begin{cases} 1, (x,y) \in \Omega^+, \\ 0, (x,y) \in \Omega^-. \end{cases}$$

The jump conditions $[\boldsymbol{u}]_\Gamma = 0, [\sigma(\boldsymbol{u}, p)\cdot\boldsymbol{n}]_\Gamma \cdot \boldsymbol{n} = 10$ are imposed at the interface Γ. In addition, the right hand is $\boldsymbol{f} = (2x, 4x)$ in this problem.

In this problem, the pressure is discontinuous and the velocity field has a kink at the interface due to different fluid viscosity. The interior nodes and boundary nodes are evenly distributed within the domain and along the boundary (see Fig. 7). The numerical and exact solutions of u_1, u_2 and p for the MBC form are shown in Figs. 8–10, respectively. We can see the numerical solutions are all in well agreements with the exact solutions.

The log-log plots of L_2 errors of the velocity and the pressure, and H_1 errors of the velocity are shown for different N_x with $m = 16$ in Fig. 11. Those figures show the errors are stable. And it is worth noting that the rate of convergence for L_2 errors of the velocity keeps fourth-order convergence. And L_2 errors of the pressure and H_1 errors of the velocity can keep the 2nd-order convergence.

Furthermore, to compare the accuracy and the numerical results between the GFDM for the MBC form and the classical PPE form, we present L_2, H_1 errors of velocity in Table 4 while L_2 errors of pressure are shown in Table 5. According to Tables 4–5, we can see the GFDM gets much less errors in e_u, $e_{u,D}$ and e_p for the MBC form than the PPE form.

5.3. Example 3: the Stokes interface problem with a hexagon interface

In the third example, we consider the Stokes interface problem on the domain $\Omega = [-1, 1] \times [-1, 1]$ with a polygon interface Γ, which can be expressed:

$$r(\theta) = \begin{cases} x(\theta) = \frac{3}{10n^2}\left(1 + 2n + n^2 - (n+1)\cos(n\theta)\right)\cos(\theta), \\ y(\theta) = \frac{3}{10n^2}\left(1 + 2n + n^2 - (n+1)\cos(n\theta)\right)\sin(\theta), \end{cases}$$

with $n = 6$ in this problem. The interior nodes and boundary nodes are evenly distributed within the domain and along the boundary (see Fig. 12). The exact solution and piecewise constant viscosity coefficient are given by Example 2.

The numerical and exact solutions of u_1, u_2 and p for the MBC form are shown in Figs. 13–15, respectively. We can see the numerical solutions are all in well agreements with the exact solutions. In Table 6, we give the L_2 error of velocity and pressure and H_1 error of velocity with $m = 16$ for Example 3. The table indicates that use the GFDM to solve

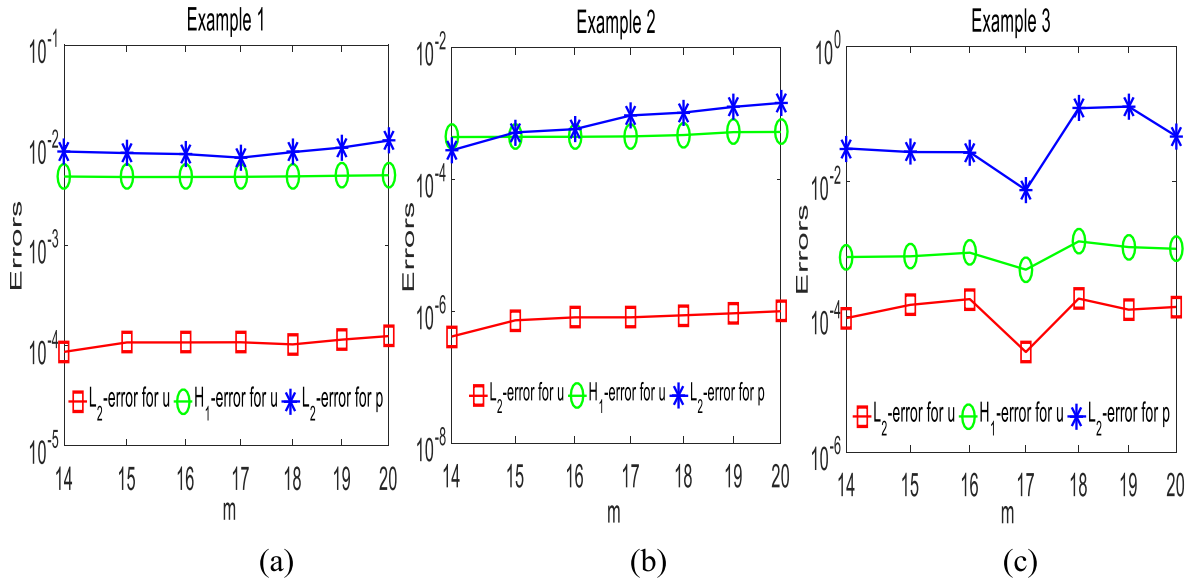


Fig. 16. The L_2, H_1 errors with different $m(m = 14, 15, 16, 17, 18, 19, 20)$ with $N_x = 40$ for (a) Example 1, (b) 2 and (c) 3.

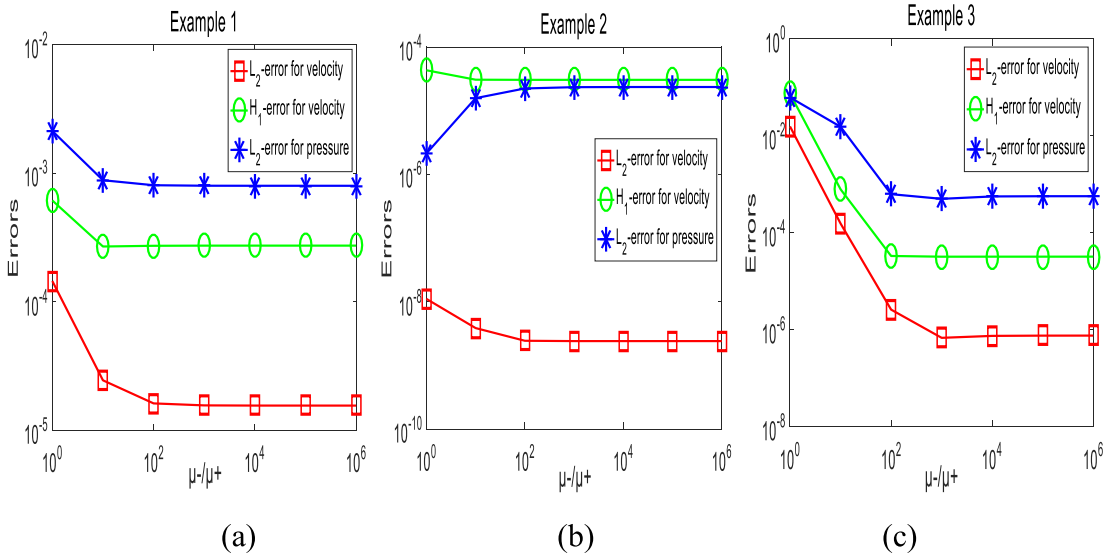


Fig. 17. Errors with different $\frac{\mu^-}{\mu^+}$ with $N_x = 200$, $m = 16$ for Example 1 to 3.

complex interface can get a relatively high numerical accuracy.

5.4. Stability test

In this section, the effect of the number of nearest nodes m of the GFDM and the jump coefficient ratio of the viscosity $\frac{\mu^-}{\mu^+}$ is tested for the above examples.

In Fig. 16, we demonstrate L_2 errors for both velocity and pressure, and H_1 errors for the velocity with different $m(m = 14, 15, 16, 17, 18, 19, 20)$and fixed $N_x = 40$. Fig. 16 (a) to (c) show that the errors computed with different$m$are relatively stable for the regular interface such as the circle in Example 1 and the line in Example 2. However, for the complex hexagon interface in Example 3, the errors have some fluctuations with the number of nearest nodes m.

Fig. 17 shows that L_2 errors for pressure, and H_1, L_2 errors for the velocity with different coefficient ratio $\frac{\mu^-}{\mu^+}$with $N_x = 200$ and $m = 16$. Here we fixed $\mu^+ = 1$ and scale μ^- according to the given ratio $\frac{\mu^-}{\mu^+}$.

Fig. 17 shows the proposed method can get accurate numerical solutions for different ratios $\frac{\mu^-}{\mu^+}$. Note that the errors decrease as the ratio of μ increases and tend stable when the ratio is over 10^2. The cause of this phenomenon is that the influence of the pressure term will gradually decrease if the μ is increasing, and that can increase the stability of the Stokes system. Moreover, the errors of the ratio equal to 1 might be not the most accurate. This is due to that the discontinuity of the pressure still leads to the existence of the interface, although the velocity is continuous in the case of ratio equal to 1.

6. Conclusions

In this paper, we apply the GFDM to solve the Stokes interface problem. To overcome the numerical pressure oscillation, we propose a new scheme to solve the Stokes interface problem. The Stokes interface

problem contains two coupled Stokes non-interface subproblems. Furthermore, the GFDM is a localized domain-type meshless method, and it can express the derivative of the unknown variable by linear summation of nearby node values, so it has advantages to handle the complex boundary conditions that appeared in the proposed scheme. In addition, the numerical results confirm the stability and accuracy of the GFDM for the new formulation in 2D cases, and the numerical examples with complex interfaces also provided convenience for our future research on practical applications of this topic.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Reference

[1] Tan ZJ, Lim KM, Khoo BC. A fast immersed interface method for solving Stokes flows on irregular domains. Comput. Fluids 2009;38:1973–83.
[2] Gross S, Reusken A. Numerical Methods for Two-Phase Incompressible Flows. New York, USA: Springer Press; 2011.
[3] Brackbill JU, Kothe D, Zemach C. A continuum method for modeling surface tension. J. Comput. Phys. 1992;100:335–53.
[4] Ausas R, Sousa F, Buscaglia G. An improved finite element space for discontinuous pressures. Comput. Methods Appl. Mech. Eng. 2010;199:1019–31.
[5] Hansbo P, Larson MG, Zahedi S. A cut finite element method for a Stokes interface problem. Appl. Numer. Math. 2014;85:90–114.
[6] Adjerid S, Chaabane N, Lin T. An immersed discontinuous finite element method for Stokes interface problems. Comput. Method. Appl. M. 2015;293:170–90.
[7] Sun P. Fictitious domain finite element method for Stokes/elliptic interface problems with jump coefficients. J. Comput. Appl. Math. 2019;356:81–97.
[8] Chessa J, Belytschko T. An extended finite element method for two-phase fluids. J. Appl. Mech. 2003;70:10–7.
[9] Reusken A. Analysis of an extended pressure finite element space for two-phase incompressible flows. Comput. Visual. Sci. 2008;11:293–305.
[10] Fries T, Belytschko T. The extended/generalized finite element method: An overview of the method and its applications. Int. J. Numer. Meth. Eng. 2010;84:253–304.
[11] Cattaneo L, Formaggia L, Iori GF, Scotti A, Zunino P. Stabilized extended finite elements for the approximation of saddle point problems with unfitted interfaces. Calcolo 2015;52:123–52.
[12] Capatina D, El-Otmany H, Graebling D, Luce R. Extension of NXFEM to nonconforming finite elements. Math. Comput. Simulat. 2017;137:226–45.
[13] Wang N, Chen J. A nonconforming Nitsche's extended finite element method for Stokes interface problems. J. Sci. Comput. 2019;81:342–74.
[14] Li Z, Lai M. The immersed interface method for the Navier–Stokes equations with singular forces. J. Comput. Phys. 2001;171:822–42.
[15] Tan Z, Lim KM, Khoo BC. An immersed interface method for Stokes flows with fixed/moving interfaces and rigid boundaries. J. Comput. Phys. 2009;228:6855–81.
[16] Hu R, Li Z. Error analysis of the immersed interface method for Stokes equations with an interface. Appl. Math. Lett. 2018;83:207–11.
[17] Alinovi E, Bottaro A. A boundary element method for Stokes flows with interfaces. J Comput. Phys. 2018;356:261–81.
[18] Hessari P. First order system least squares method for the interface problem of the Stokes equations. Comput Math with Appl 2014;68:309–24.
[19] Hessari P. Least squares spectral method for the two-dimensional Stokes interface problems. J. Comput. Appl. Math. 2020;364:112325.
[20] Ahmad M, Islam S, Ullah B. Local radial basis function collocation method for stokes equations with interface conditions. Eng. Anal. Bound. Elem. 2020;119:246–56.
[21] Liszka T, Orkisz J. The finite difference method at arbitrary irregular grids and its application in applied mechanics. Comput . Struct. 1980;11:83–95.
[22] Liszka T. An interpolation method for an irregular net of nodes. Int. J. Numer. Methods Eng. 1984;20:1599–612.
[23] Benito JJ, Ureña F, Gavete L. Influence of several factors in the generalized finite difference method. Appl. Math. Model. 2001;25:1039–53.
[24] Gavete L, Gavete ML, Benito JJ. Improvements of generalized finite difference method and comparison with other meshless method. Appl. Math. Model. 2003;27:831–47.
[25] Benito JJ, Urena F, Gavete L, Alvarez R. An h-adaptive method in the generalized finite differences. Comput. Meth. Appl. Mech. Eng. 2003;192:735–59.
[26] Gavete L, Benito JJ, Ureña F. Generalized finite differences for solving 3D elliptic and parabolic equations. Appl. Math. Model. 2016;40:955–65.
[27] Gavete L, Ureña F, Benito JJ, García A, Ureña M, Salete E. Solving second order non-linear elliptic partial differential equations using generalized finite difference method. J. Comput. Appl. Math. 2017;318:378–87.
[28] Fan CM, Li PW, Yeih WC. Generalized finite difference method for solving two-dimensional inverse Cauchy problems. Inverse. Probl. Sci. Eng. 2015;23(5):737–59.
[29] Gu Y, Wang L, Chen W, Zhang CZ, He XQ. Application of the meshless generalized finite difference method to inverse heat source problems. Int. J. Heat. Mass. Transf. 2017;108:721–9.
[30] Xing YN, Song LN, He XM, Qiu CX. A generalized finite difference method for solving elliptic interface problems. Math. Comput. Simulat. 2020;178:109–24.
[31] Li PW, Fu ZJ, Gu Y, Song LN. The generalized finite difference method for the inverse Cauchy problem in two-dimensional isotropic linear elasticity. Int. J. Solids. Struct. 2019;174-175:69–84.
[32] Hidayat MIP, Widyastuti RFajarinF. A meshless generalized finite difference method for 2D elasticity problems. Eng. Anal. Bound. Elem. 2020;117:89–103.
[33] Song LN, Li PW, Gu Y, Fan CM. Generalized finite difference method for solving stationary 2D and 3D stokes equations with a mixed boundary condition. Comput. Math. Appl. 2020;80:1726–43.
[34] Li PW, Chen W, Fu ZJ, Fan CM. Generalized finite difference method for solving the double-diffusive natural convection in fluid-saturated porous media. Eng. Anal. Bound. Elem. 2018;95:175–86.
[35] Li PW, Fan CM. The generalized finite difference method for two-dimensional shallow water equations. Eng. Anal. Bound. Elem. 2017;80:58–71.
[36] Gu Y, Lei J, Fan CM, He XQ. The generalized finite difference method for an inverse time-dependent source problem associated with three-dimensional heat equation. Eng. Anal. Bound. Elem. 2018;91:73–81.
[37] Li PW. Space-time generalized finite difference nonlinear model for solving unsteady Burgers' equations. Appl. Math. Lett. 2021;114:106896.
[38] Gavete L, Urena F, Benito JJ, Salete E. A note on the dynamic analysis using the generalized finite difference method. J. Comput. Appl. Math. 2013;252:132–47.
[39] Chan HF, Fan CM, Kuo CW. Generalized finite difference method for solving two dimensional non-linear obstacle problems. Eng. Anal. Bound. Elem. 2013;37:1189–96.